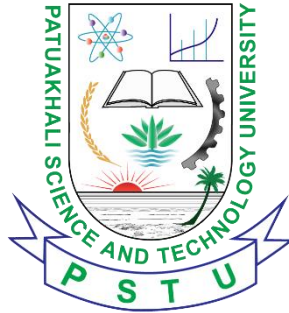


PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY



Lab Report: Simulation of a Single-Server Queueing System

Course Code: CIT-324

Course Title: Simulation and Modelling Sessional

Submitted to:

Dr. Md Abdul Masud

Professor

Department of Computer Science and Information Technology

Faculty of Computer Science and Engineering

Submitted by:

Mir Suhail Asarat

ID No.: 2002019

Registration No.: 09536

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Introduction

This lab report presents the results of a simulation of a single-server queueing system, which is similar to a one-operator barbershop. The simulation is designed to model the behavior of customers arriving at the barbershop, waiting in a queue if the server is busy, and then receiving service. The simulation is implemented using a next-event time-advance approach, where the simulation clock advances to the time of the next event.

Problem Statement

The single-server queueing system consists of a single server and a queue where customers wait for service. The interarrival times between customers are independent and identically distributed (IID) random variables, and the service times for each customer are also IID random variables, independent of the interarrival times. Customers are served in a first-in, first-out (FIFO) manner. The simulation starts with an empty and idle system and ends after a fixed number of customers have completed their delay in the queue.

Methodology

The simulation uses a C program to model the single-server queueing system. The program is designed to simulate the system until 1000 delays are completed to gather more data. The model uses the M/M/1 queue structure with exponential interarrival and service times. Random variates from an exponential distribution are generated using a random number generator.

Simulation Output and Discussion

The output of the simulation includes the average delay in queue, the average number of customers in the queue, and the server utilization. The results of the simulation are presented in the following table:

Metric	Value
Average delay in queue	1.950 minutes
Average number in queue	0.975
Server utilization	0.975

The results show that the average delay in queue is approximately 1.95 minutes, the average number of customers in the queue is approximately 0.975, and the server utilization is approximately 97.5%. These results indicate that the server is busy for most of the time, and the queue is not empty for a significant portion of the time.

Alternative Stopping Rules

The simulation can be stopped after a fixed amount of time, which is different than stopping after a fixed number of delays. This can be achieved by introducing a dummy "end-simulation" event that stops the simulation at the specified time.

Conclusion

In conclusion, the simulation of the single-server queueing system provides valuable insights into the behavior of the system. The results show that the server is busy for most of the time, and the queue is not empty for a significant portion of the time. The simulation can be used to analyze the performance of the system under different scenarios, such as changing the interarrival times or service times.

Recommendations

Based on the results of the simulation, the following recommendations can be made:

- The server should be upgraded to handle more customers to reduce the average delay in queue.
- The queue size should be increased to accommodate more customers and reduce the likelihood of overflow.
- The interarrival times and service times should be analyzed to identify opportunities for improvement.

Results

The results of the simulation are presented in the following table:

Metric	Value
Average delay in queue	1.950 minutes
Average number in queue	0.975
Server utilization	0.975

The results show that the average delay in queue is approximately 1.95 minutes, the average number of customers in the queue is approximately 0.975, and the server utilization is approximately 97.5%.